Leah Hoogstra

Curriculum Vitae

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https://leahhoogstra.com

Research Experience

PhD Candidate in Applied Mathematics *Sep 2025 - present*
Brown University

- Current project, advised by Dr. Baylor Fox-Kemper and Dr. Mara Freilich: Developing machine-learning models to predict sea level at NOAA tide gauges in Narragansett Bay and surrounding waters, providing fast, flexible alternatives to computationally intensive physics-based models. The work benchmarks ML predictions against the Ocean State Ocean Model (a ROMS model) and existing tidal predictions, with applications to flood forecasting, climate scenario evaluation, and transferability to other coastal regions.
- Project in development, advised by Dr. Baylor Fox-Kemper and Dr. Mara Freilich: Investigating the physical mechanisms that control microplastic transport in tidal estuaries, with a focus on how wave–mean flow interactions modify particle pathways.This theoretical work will be grounded in Narragansett Bay, connecting to particle-tracking experiments in the Ocean State Ocean Model and field sampling data from collaborating researchers.
- Completed project, advised by Dr. Björn Sandstede: Developed a probabilistic mathematical framework for multidimensional demographic diversity metrics, formulating and proving a theorem establishing anti-correlation and tight bounds between "intersecting diversity" and "shared identity," with applications to empirical social datasets.

Geophysical Fluid Dynamics Fellow *Jun 2026 - Aug 2026*
Woods Hole Oceanographic Institution

- Ongoing project, advised by Dr. Jim Thomas: Using asymptotics to derive a reduced model that captures interactions between a broad spectrum of internal waves and a geostrophically balanced eddy field via a discrete set of coupled wave amplitude equations. We are working to validate the reduced model by comparing it with numerical simulations of the non-hydrostatic Boussinesq equations.

Master's Student in Mathematics *Sep 2021 - May 2023*
California Polytechnic State University, San Luis Obispo

- Thesis, advised by Dr. Paul Choboter and Dr. Ryan K. Walter: Applied Lagrangian coherent structure (LCS) theory to evaluate ocean surface circulation in a Central California offshore Wind Energy Area, comparing FTLE- and geodesic-based hyperbolic LCS detection using HF radar data.
- Summer research, advised by Dr. Charles D. Camp: Led a team of three undergraduates, researching effects of periodic and quasi-periodic forcing on the pace of ice ages in five coupled, nonlinear ODE models of the late Pleistocene climate.

Education

Brown University
Ph.D. Applied Mathematics *expected May 2028*

- Advisors: Dr. Baylor Fox-Kemper and Dr. Mara Freilich
- Co-organizer of the Social Equity and Applied Mathematics (SEAM) Seminar [website]

California Polytechnic State University, San Luis Obispo
M.S., Mathematics, *summa cum laude* *June 2023*

- Advisors: Dr. Paul Choboter and Dr. Ryan K. Walter
- Thesis: *Detecting coherent transport structures in ocean surface flows*

Calvin College (now Calvin University)
B.A., Mathematics and International Development *May 2013*

Scholarship

Publications

Walter, R.K., Choboter, P., Rocha Jayasinha, A.E., Coutin, R., Luoma, C., **Hoogstra, L.**, Pratt, C., Idriss, M., Ehrnstein, E., and Robbins, I., Seasonal patterns of surface currents and Lagrangian coherent structures from a decade of high-frequency radar observations in a central California offshore wind energy area, *Regional Studies in Marine Science*, 103, 105566, 2026. [doi]

Hoogstra, L., Slyman, K., and Sandstede, B., Quantitative metrics for trait and identity distributions, arXiv preprint, 2025. [preprint]

Hoogstra, L., Detecting coherent transport structures in ocean surface flows, Master's thesis, California Polytechnic State University, San Luis Obispo, 2023. [thesis]

Presentations

Hoogstra, L., Huang, X., Freilich, M., and Fox-Kemper, B., Machine learning methods for storm surge bias correction in mid-latitude coastal regions: A Narragansett Bay case study, Poster presentation, AGU Annual Meeting, San Francisco, CA, December 2026.

Hoogstra, L., Ehrnstein, E., Choboter, P., and Walter, R.K., Spatially and temporally coherent transport structures in the Central California offshore Wind Energy Area, Video submission, Gallery of Fluid Motion, 76th Annual Meeting of the APS Division of Fluid Dynamics, November 2023. [doi]

Hoogstra, L., Ehrnstein, E., Choboter, P., and Walter, R.K., Spatially and temporally coherent transport structures in the Central California offshore Wind Energy Area, Poster presentation, College of Science and Mathematics Research Conference, California Polytechnic State University, San Luis Obispo, October 2023.

Hoogstra, L., Forced dynamical systems in paleoclimate modeling, Invited talk, American Association of University Women, San Luis Obispo Chapter, January 2023.

Coutin, R., Fletcher, C., **Hoogstra, L.**, Stensland, L., and Camp, C.D., Forced dynamical systems in paleoclimate modeling, Poster presentation, Frost Undergraduate Research Poster Symposium, California Polytechnic State University, San Luis Obispo, October 2022.

Relevant Graduate Coursework

Geophysical Fluid Dynamics	Computational Fluid Dynamics	Fluid Mechanics I
Research in Geological Science	Mathematics and Climate	Climate Modeling I
Nonlinear Dynamical Systems I	Numerical Solutions of PDEs I	Machine Learning
Nonlinear Dynamical Systems II	Real Analysis	Functional Analysis

Work Experience

Reporting Analyst II *Jul 2019 - Aug 2021*
Fidelity National Information Services (FIS)

- Automated and optimized revenue reporting processes for a large financial services firm, compiling over 2 million rows of data quarterly with Python and building interactive Power BI reports used by 200+ employees, including executives, to analyze client, product, and location trends.

Refugee Program Data Specialist *Aug 2018 - Jul 2019*
Mennonite Central Committee SALT Program - placed at Collateral Repair Project

- Managed data systems on site for a nonprofit serving refugees in Amman, Jordan, including transitioning the primary database to Azure SQL and creating grant-tracking and family-registration applications in Salesforce and AppSheet. Synthesized siloed data for 5,000+ families into a cohesive architecture to support program operations, case management, and sustainable long-term technical planning.

Research Associate *May 2014 - Jun 2017*
Basis Policy Research

- Supported a variety of applied K–12 education policy projects for private foundations, school districts, charter networks, and state governments. Leveraged technical skills in SQL, Stata, Tableau, Power BI, and R to manage large student datasets, build dashboards and data pipelines, implement a student-teacher matching algorithm, and generate analyses and visualizations that directly informed client decision-making.

Research Coordinator *Feb 2012 - Apr 2014*
Calvin College Center for Social Research (now DataWise Consulting)

- Supported a wide range of social science research initiatives by integrating multi-year survey and registrar data, creating analytic datasets, and designing interactive Tableau reports to inform departmental decisions. Designed and administered online Qualtrics surveys, provided technical support, and co-led staff training in database theory and Microsoft Access.

Technical Skills

Python (PyTorch, scikit-learn, pandas, numpy, scipy, xarray, statsmodels, matplotlib, seaborn, cartopy, netCDF4, requests), LaTeX, Matlab, SQL, R, Tableau, Power BI